Session with Harshit Sir



Class X • Chapter 7

Coordinate Geometry: The Ultimate Challenge

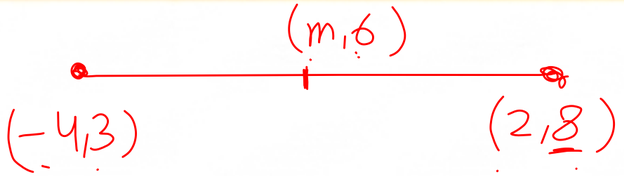
Conquering the hardest HOTS questions in the syllabus.

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Complex Section Formula

Find the ratio in which the point P(m, 6) divides the line segment joining the points A(–4, 3) and B(2, 8). Also find the value of m.

Soln



$$y = \frac{my_2 + ny_1}{m+n}$$

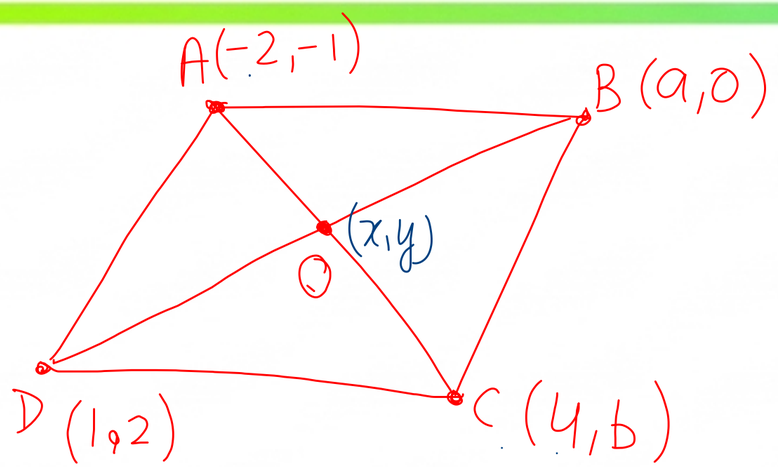
$$6 = \frac{m(8) + n(3)}{m+n}$$

Ans: Ratio = 3: 2 | m = –2/5

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Mid-Point Logic

If the points A(−2, −1), B(a, 0), C(4, b) and D(1, 2) are the vertices of a parallelogram, taken in order, find the values of a and b.



Mid-point of AC

$x = \frac{-2+4}{2} = 1$

$y = \frac{-1+b}{2}$

$\frac{a+1}{2} = 1$

$a+1 = 2$

$a = 1$

Mid-point of BD

$x = \frac{a+1}{2}$, $y = 1$

$\frac{-1+b}{2} = 1$

$-1+b = 2$

$b = 3$

Ans: a = 1, b = 3

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Qn. If the mid-point of the line joining $(3,4)$ and $(k,7)$ is (x,y) and $2x+2y+1=0$, find the value of k.

$$x = \frac{3+k}{2} \qquad y = \frac{7+4}{2}$$

$$= \frac{11}{2}$$

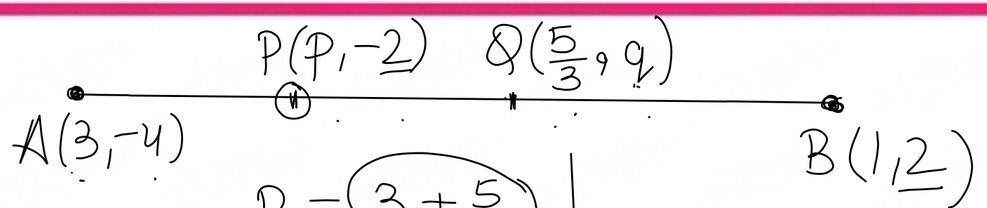
$$2\left(\frac{3+k}{2}\right) + 2\left(\frac{11}{2}\right) + 1 = 0$$

$$\Rightarrow 3+k+11+1=0$$

$$\Rightarrow k+15=0$$

$$\Rightarrow k=-15$$

The line segment joining the points (3, –4) and (1, 2) is trisected at the points P(p, –2) and Q(5/3, q). Find the values of p and q.



$$p = \frac{3 + \frac{5}{3}}{2}$$

$$= \frac{\frac{14}{3}}{2}$$

$$= \frac{14}{6}$$

$$= \frac{7}{3}$$

$$q = \frac{-2 + 2}{2}$$

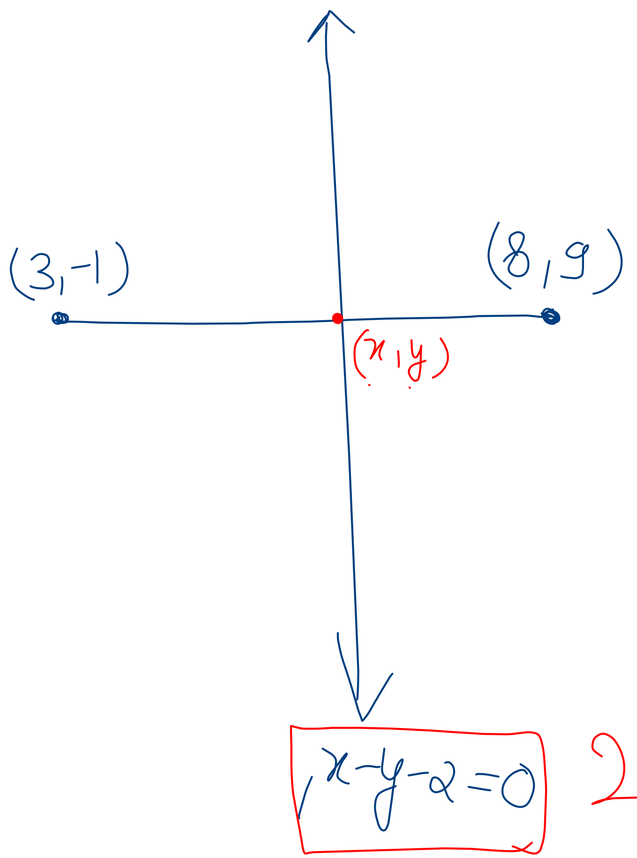
$$= 0$$

Ans: p = 7/3, q = 0

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Qn. Determine the ratio in which straight line $x-y-2=0$ divides the line segment joining the points $(3,-1)$ and $(8,9)$

$x-y=2$



Section Formula

$$x=\frac{m(8)+n(3)}{m+n} \qquad y=\frac{m(9)+(n)-1}{m+n}$$

$$x-y=\frac{8m+3n}{m+n}-\left(\frac{9m-1n}{m+n}\right)$$

$$x-y=\frac{8m+3n-9m+1n}{m+n}$$

$$2m+2n=\frac{-1m+4n}{1} \quad \Rightarrow \quad 3m=2n$$

$$\frac{m}{n}=\frac{2}{3}$$

$m:n=2:3$

Equidistance Challenge

If Q(0, 1) is equidistant from P(5, –3) and R(x, 6), find the values of x. Also find the distances QR and PR.

Ans: $x = \pm 4$ | $QR = \sqrt{41}$ | $PR = \sqrt{82}$ or $9\sqrt{2}$

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Area & Collinearity

Find the value of m for which the points with coordinates (3, 5), (m, 6) and (1/2, 15/2) are collinear.

Ans: m = 2

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Line Intersection Ratio

Find the ratio in which the line $2x + y - 4 = 0$ divides the line segment joining the points $A(2, -2)$ and $B(3, 7)$.

Ans: Ratio = 2:9 internally

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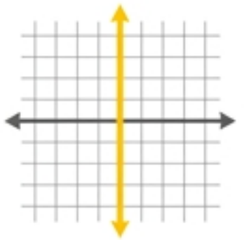
Variable Collinearity

For what values of k are the points (k, 2 – 2k), (–k + 1, 2k) and (–4 – k, 6 – 2k) collinear?

Ans: k = 1/2 or -1

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The Ordinate Anchor



Find the ratio in which the point P whose ordinate is —3 divides the join of A(–2, 3) and B(5, —15/2). Hence find the coordinates of P.

Ans: Ratio = 4:3 | P(2, -3)

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